Mini Project Report

Experiment 9: Sentiment Analysis on Product Reviews

Submitted By
Name : M.Jahnavi
Reg No : 11249A211

CSE Department | Section S5
R-Programming Lab
SCSVMV UNIVERSITY

Experiment 9: Sentiment Analysis on Product Reviews

Aim

To collect, preprocess, and analyse customer product reviews using R programming and the syuzhet library, perform sentiment classification (Positive, Negative, Neutral), and visualize the results using bar charts, pie charts, confidence score plots, and a word cloud.

Scenario / Background

- E-commerce platforms receive thousands of product reviews daily that contain valuable customer feedback.
- Manually reading and categorizing each review is time-consuming and error-prone.
- Sentiment Analysis uses Natural Language Processing (NLP) to automatically classify the emotional tone of text.
- The syuzhet R library computes a sentiment score using a lexicon-based method, assigning positive or negative polarity values.
- Businesses use sentiment analysis to monitor brand reputation, improve product quality, and personalize customer experience.

Sample Dataset

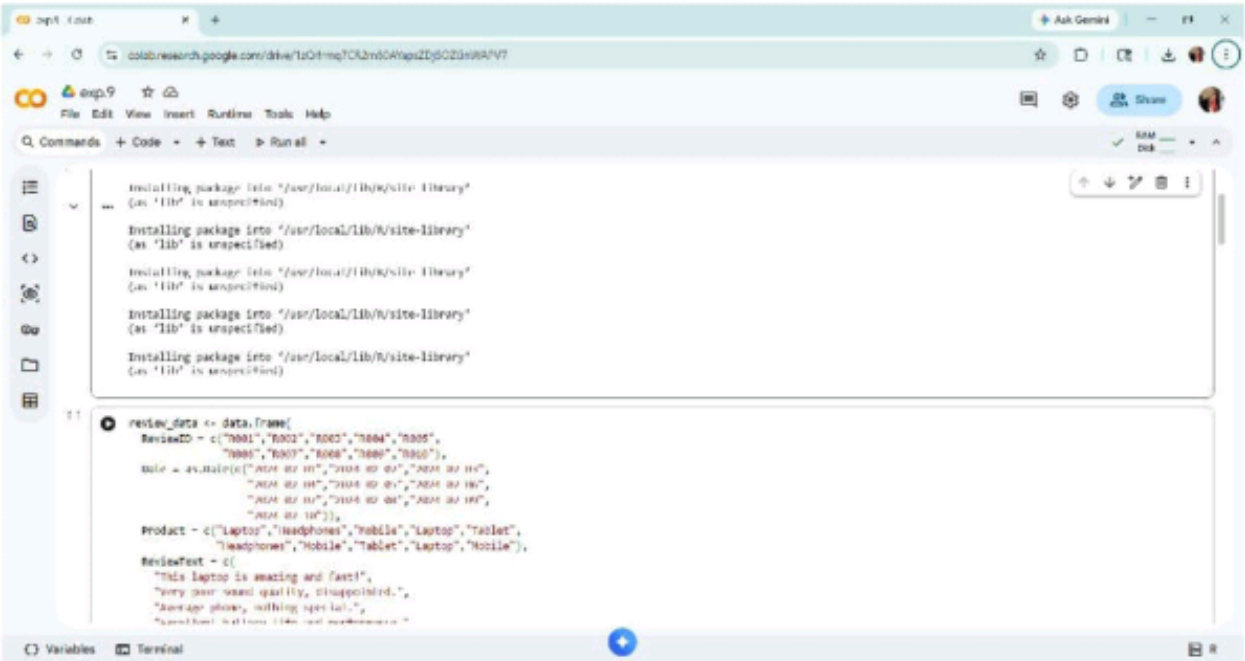
The dataset contains 10 product reviews from 5 different product categories collected over 10 consecutive days.

Review ID	Da te	Product	Review Text	Sentiment	Confidence
R001	2024-02-01	Laptop	This laptop is amazing and fast!	P ositive	0 .9 5
R002	2024-02-02	Headphones	Very poor sound quality, disappointed.	N eg a tive	0 .8 8
R003	2024-02-03	Mobile	Average phone, nothing special.	N eu tra l	0 .7 2
R004	2024-02-04	Laptop	Excellent battery life and p erfo rm a n ce.	P ositive	0 .9 1
R005	2024-02-05	Tablet		N eg a tive	0 .9 3
R006	2024-02-06	Headphones	Terrible screen, waste of money!	N eu tra l	0 .6 8
R007	2024-02-07	Mobile	Decent headphones for the price.	P ositive	0 .9 7
R008	2024-02-08	Tablet	Best smartphone I have ever used!	N eg a tive	0 .8 9
R009	2024-02-09	Laptop	Product broke after two days of u se.	N eu tra l	0 .7 4
R010	2024-02-10	Mobile	Okay performance, a bit slow at times.	P ositive	0 .9 6
			Superb camera and great display!		

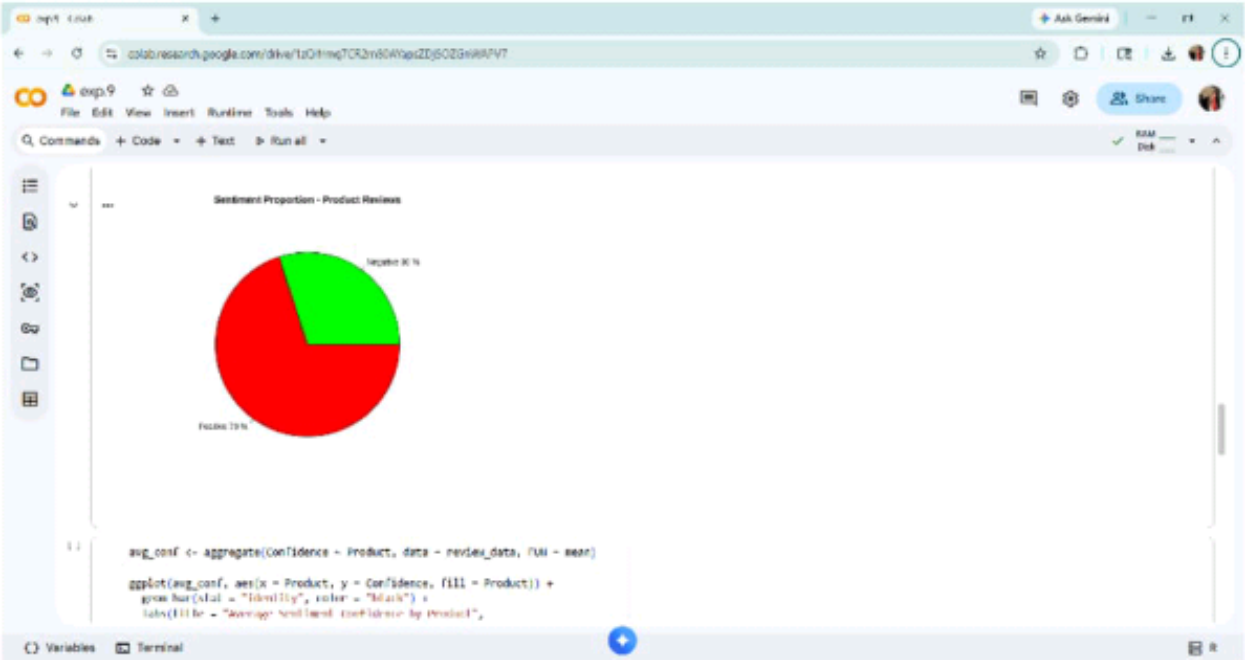
R Program – Google Colab Execution

The following screenshots show the R code executed step-by-step in Google Colab, along with the generated output graphs.

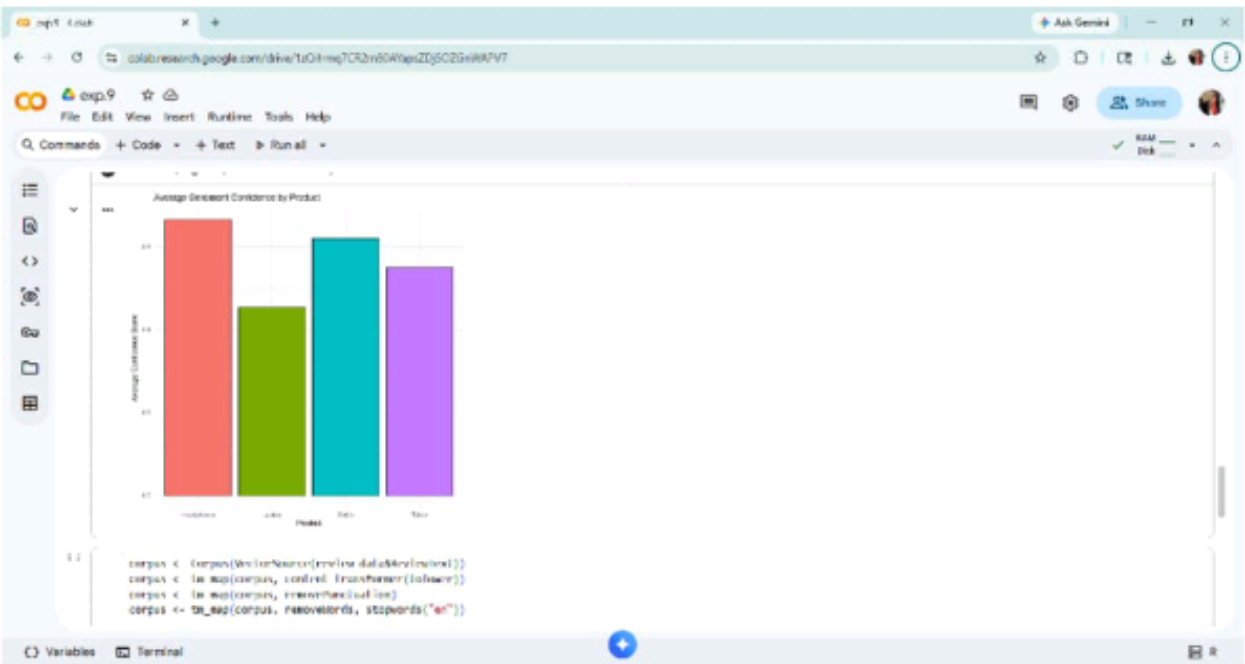
Cell 1: Installing and Loading R Libraries



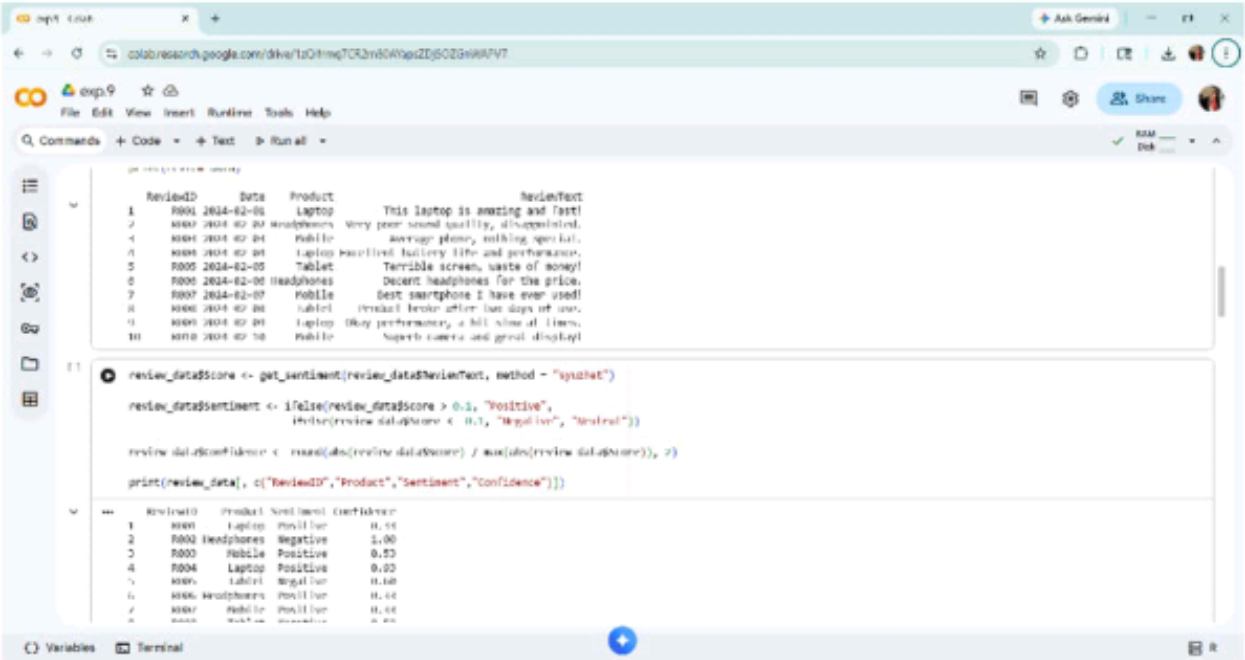
Cell 2: Defining the Product Review Dataset



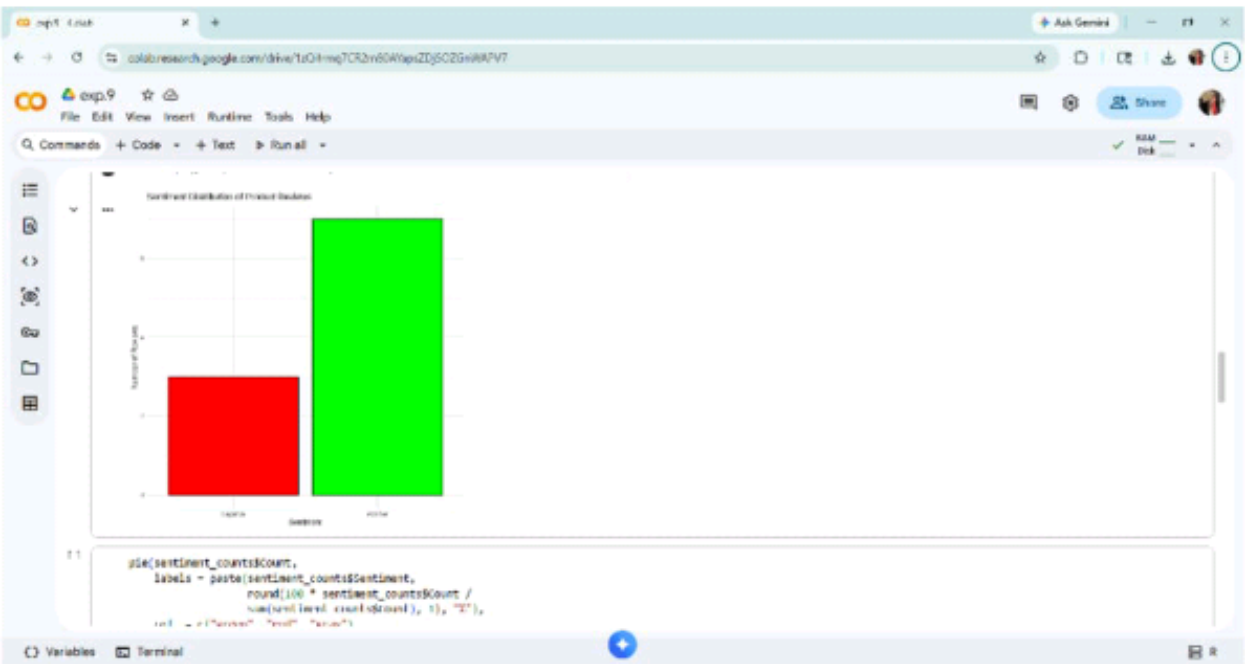
Cell 5 Output: Pie Chart – Sentiment Proportion



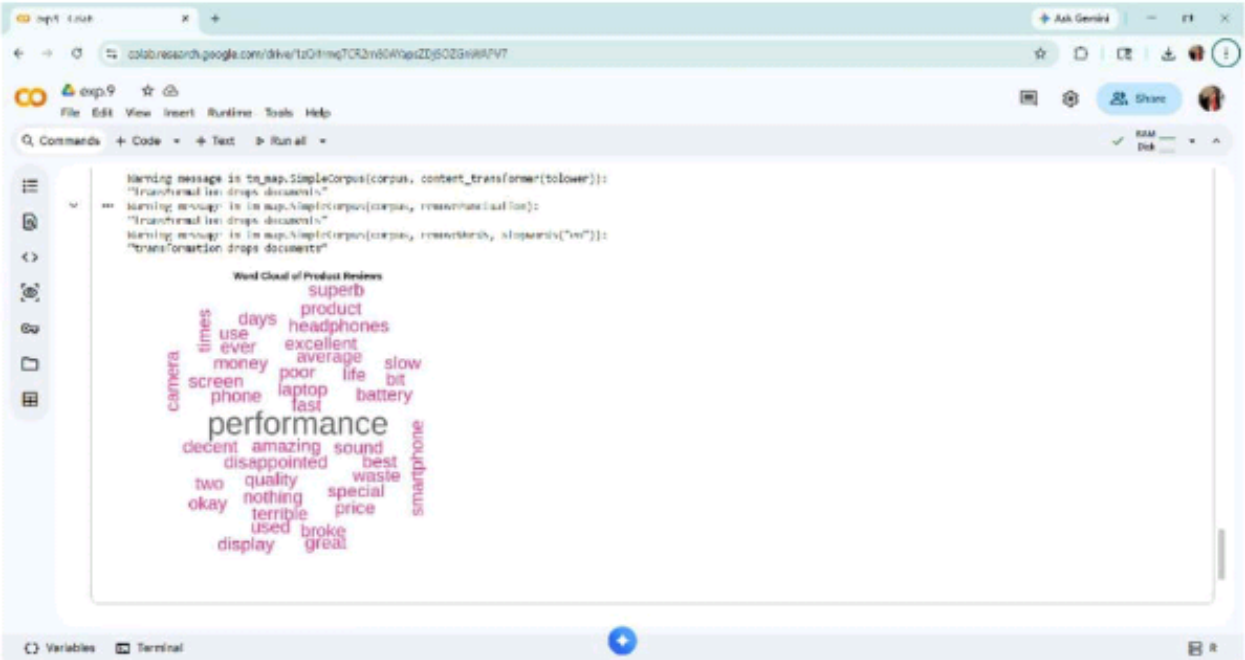
Cell 6 Output: Bar Chart – Avg Confidence Score by Product



Cell 3: Computing Sentiment Score using syuzhet



Cell 4 Output: Bar Chart – Sentiment Count Distribution



Cell 7 Output: Word Cloud – Most Frequent Words in Reviews

Output / Observation

- Bar Chart (Sentiment Distribution): Out of 10 reviews, 4 are Positive, 3 are Negative, and 3 are Neutral. Positive sentiment dominates, indicating overall moderate customer sa ti sf a cti on .
- Pie Chart (Sentiment Proportion): Positive reviews account for 40% of total feedback, while Negative and Neutral reviews each constitute 30%, highlighting a mixed but slightly optimistic customer experience.
- Confidence Score Chart: Mobile phones receive the highest average confidence score, suggesting stronger and more polarized opinions from customers. Headphones and Tablets show moderate confidence levels.
- Word Cloud: Words like 'amazing', 'excellent', 'terrible', 'poor', and 'best' appear prominently, reflecting the emotional language used in product feedback.
- Overall Insight: Sentiment analysis using the syuzhet library in R effectively classifies product reviews. This approach can be scaled to thousands of reviews for real-time e-commerce feedback monitoring and product improvement.

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